Paleomagnetism of the Uairen Formation, Roraima Group, southeastern Venezuela: evidence for one of the oldest (Middle Proterozoic) depositional remanent magnetizations

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This study reports paleomagnetic results from nine localities (22 sites) of the Uairen Formation, the basal and oldest unit of the Roraima sedimentary sequence (Roraima Group) in the Precambrian Guayana shield (southeastern Venezuela). Three distinct natural remanent magnetizations (NRM's), U1N, U1R, and U2, were isolated by alternating-field and thermal demagnetization. U1N and U1R NRM's have mean directions $D = 342°$, $I = 6°$ ($k = 14$, $\alpha_{95} = 10°$, $N = 14$ sites) and $D = 155°$, $I = 10°$ ($k = 8$, $\alpha_{95} = 17°$, $N = 8$ sites), respectively. Because of their high coercivities (H_C) and high (>580°C) laboratory unblocking temperatures (T_{UB}), it is believed that these NRM's are carried mainly by fine-grained magnetite and (or) hematite. The U2 NRM has $D = 338°$, $I = -7°$ ($k = 5$, $\alpha_{95} = 25°$, $N = 6$ sites). U2 is a low-H_C (20 mT) and low-T_{UB} (<300°C) NRM carried by coarse-grained magnetite. Despite its low Fisher's k value, the U2 grand mean closely resembles that for U1N. Directional similarities between U1N, U1R, and U2 to groups I and II previously reported for diabase dikes and sills that intruded the Roraima sedimentary sequence (Roraima Intrusive Suite) suggest that primary NRM's of these sediments have been completely overprinted by thermal and hydrothermal effects associated with the intrusive suite. The apparent absence of diabase intrusions at the sampled localities, but mainly the low metamorphic grade of the Uairen samples revealed by thin-section analyses, lead to the alternative conclusion that U2 is a primary detrital remanent magnetization (DRM) dating from the time of deposition of the Uairen sediments, and U1N and U1R are later thermochemical overprints (TCRM's) acquired shortly after deposition by oxidation of fine-grained magnetite. Directional scatter of U2 could be explained as the consequence of misaligning effects of mechanic over magnetic torques acting on coarse-grained magnetite. An age between 1580 and 1960 Ma, derived from radiometric constraints, is suggested for the Uairen components. Thus U2 would be one of the oldest DRM's ever reported.

Les résultats d'une étude paléomagnétique sont présentés pour neuf localités (22 sites) de la Formation d'Uairen, la plus vieille unité de la base de la séquence sédimentaire de Roraima (Groupe de Roraima) du Bouclier précambrien des Guyanes (région sud-est du Venezuela). La désaimantation par champ alternatif et par chauffage a permis d'isoler trois aimantations rémanentes naturelles (ARN) distinctes : U1N, U1R et U2. Les ARN U1N et U1R fournissent les directions moyennes de $D = 342°$, $I = 6°$ ($k = 14$, $\alpha_{95} = 10°$, $N = 14$ sites) et $D = 155°$, $I = 10°$ ($k = 8$, $\alpha_{95} = 17°$, $N = 8$ sites), respectivement. Les fortes coercivités (H_C) et les hautes températures de déblocage (>580°C) mesurées en laboratoire révèlent que les ARN sont portées principalement par la magnétite à grain fin et (ou) l'hématite. L'ARN U2 fournit une direction $D = 338°$, $I = -7°$ ($k = 5$, $\alpha_{95} = 25°$, $N = 6$ sites). U2 est une ARN caractérisée par H_C faible (20 mT) et par une température de déblocage basse (<300°C), et elle est portée par la magnétite à grain grossier. Même si le paramètre k de Fisher est petit, la grande moyenne de U2 et de U1N diffèrent peu l'une de l'autre. Les directions similaires de U1N, U1R et U2 des groupes I et II connues pour les dykes et les filons-couches qui recoupent la séquence sédimentaire de Roraima (suite intrusive de Roraima) suggèrent que les ARN primaires de ces sédiments furent entièrement surimprimées par les effets thermiques et hydrothermaux associés à la suite intrusive. L'absence apparente d'intrusions de diabase aux localités échantillonnées, mais surtout le faible degré de métamorphisme des échantillons d'Uairen révélé par l'étude des lames minces, plaident en faveur d'une autre conclusion invoquant que U2 est une aimantation rémanente détritique primaire (ARD), acquise durant le stade de dépôt des sédiments d'Uairen, et que U1N et U1R représentent des surimpressions thermochimiques plus tardives dues à une oxydation de la magnétite à grain fin peu de temps après le dépôt. La dispersion des directions de U2 peut être expliquée comme le résultat d'un désalignement réalisé mécaniquement par les torques magnétiques qui agissaient sur la magnétite à grain grossier. Un âge compris entre 1580 et 1960 Ma, déduit de contraintes radiométriques, est attribué aux composantes de l'Uairen. Alors, U2 serait l'une des plus anciennes ARD signalée jusqu'à présent.

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Introduction

The Roraima Group is a 1800 m thick sedimentary sequence in the Guayana Precambrian shield (southeastern Venezuela) (Reid 1972). The age range of this group is controversial. Stipp and Teixeira (1975) and Ghosh (1977) have pointed out that at different locations these sediments must have different ages despite their lithological similarities. They also have suggested that such similarities could be due to the persistence of a single tectonic sedimentary enviroment throughout hundreds of millions of years.

The sedimentary sequence has been intruded at different levels by a network of diabase dikes and sills known as the Roraima Intrusive Suite (Bellizia 1957; Hawkes 1966). Paleomagnetic studies in samples from the Roraima Group have focused mainly on this intrusive suite (Hargraves 1968; Veldkamp et al. 1971; Onstott and Hargraves 1981). A few paleomagnetic results of the Roraima sandstones have led Onstott and Hargraves (1981) to the conclusion that these samples have been chemically remagnetized by pervasive hydrothermal alteration.

We report here the first paleomagnetic results of the Uairen Formation, the basal and oldest part of the Roraima Group (Reid 1972), with a thickness of about 850 m. The purpose of this study is to isolate a detrital remanent magnetization (DRM) and to estimate its possible age. DRM is the natural remanent magnetization (NRM) of most sedimentary rocks

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and is acquired by the rotation and alignment of magnetic particles in the presence of the geomagnetic field during and (or) shortly after deposition. This type of NRM has been extensively used in paleomagnetic and magnetostratigraphic studies of Phanerozoic sediments, but only in few cases has it been recognized in Precambrian rocks (e.g., Henthorn 1981; Zhang and Zhang 1985).

Geological setting and sampling

The Roraima Province comprises a large unit of Precambrian sediments that have been intruded by diabase dikes and sills of different ages. These sediments extend over an area of about 400 000 km^2 extending from Tafelberg (Suriname) in the east to the Macarena Ridge (Colombia) in the west, and from the Orinoco River (Venezuela) in the north to the Branco River (Brazil) in the south (Gonzalez de Juana et al. 1980).

Roraima sediments are mostly of continental origin, consisting of conglomerates, feldspathic sandstones, quartz, quartzite, siltstones, cherts, shales, arkoses, and jasper (Bellizia 1957). Geomorphologically these sediments show prominent ridges that have been eroded to form tabular hills known as tepuis.

In large areas of the province the bedding lies near horizontal or dips a few degrees to the north. Yanez (1977) has reported an average inclination between 8 and 12° for this unit. The source of Roraima sediments is controversial. Mendoza (1977), on the basis of a study of the siliceous and cross-bedded characteristics of these sediments, has pointed out that the source could be located north of latitude 6°30′N. Ghosh (1977), studying paleocurrent directions, suggested that these sediments were derived from the south and east, that their source was cratonic, and that they were deposited largely in a shallow marine enviroment.

The stratigraphic sequence of the Roraima province has been studied by Aguerrevere et al. (1939), Yanez (1969, 1977), Reid (1972), Keats (1977), and Ghosh (1977), among others. Reid (1972) has recognized four formations for this group in Venezuela, namely (from base to top) Uairen, Cuquenan, Uaimapue, and Mataui.

Since the Roraima Group has been intruded by sills and dikes of diabase at different levels, and because the abundance and development of these towards the north and in adjacent areas of the Guayana shield, Bellizia (1957) suggested the existence of a zone of weakness of great magnitude at this locality.

The age of the Roraima Group is largely unconstrained. Hawkes (1966) distinguished within the Roraima Province a group of younger dikes and sills (minor dike suite) of Permo-Triassic age (221 Ma) and a major intrusive suite of dikes and sills (Roraima Intrusive Suite) of Precambrian age. McDougall et al. (1963) working on diabases from Low Kopinang and Kamarang (Guyana) reported Rb/Sr determinations in the range of 1540–2090 Ma and established an older limit for Roraima of 2100 Ma. Snelling and McConnell (1969) rejected this limit and proposed a maximum Rb/Sr age of 1965 Ma for the Kopinang diabases, but they accepted the possibility of younger intrusions with ages of up to 1850 Ma. Priem et al. (1973) reported a Rb/Sr age of 1599 Ma for acidic pyroclastic rocks from Tafelberg (Suriname) and 1450–1650 Ma for acidic intrusives from Surucu and Surumo (Brazil). The Tafelberg pyroclastics have been correlated, in the stratigraphic columns of the Roraima Group, to the Uaimaipue Formation (Venezuela) on top of Uairen (Gonzalez de Juana et al. 1980). Rb/Sr age determinations on ash tuffs and pyroclastic rocks interbedded with sandstones of the Roraima Group in Venezuela are 1700 Ma (Gaudette and Olszewski 1981) and 1580 Ma (Pringle and Teggin 1985), respectively.

In the Santa Elena area, where this study was carried out, the Cuchivero Province forms the basement on which Roraima sediments lie uncorformably. Rb/Sr and K/Ar ages of volcanic rocks in the Cuchivero Province range between 1640 and 1960 Ma. However, most of these ages are concentrated in a narrower interval from 1810 to 1870 Ma (e.g., Snelling and Berrangé 1970; Stipp and Teixeira 1975; Amaral and Halpern 1975; Gaudette et al. 1978).

For this study nine localities were sampled in the Uairen formation, the lowest and oldest unit within the Roraima Group (Fig. 1). The distribution of these localities was dictated by the availability of fresh exposures from highway cuts and rivers. A total of 133 separately oriented samples, which included drill cores and hand samples, were collected from 22 sites (about 6 samples per site). Sites are understood here as sedimentary horizons separated from each other by a minimum of a few centimetres (e.g., locality 1) and a maximum of 2 m (e.g., locality 6).

The origin of Uairen sediments is believed to be fluvial. They comprise conglomerates, sandstones, quartzites, quartzose sandstones, and indurated sandstones. Intraformational conglomerates are common in the lowest part of the formation. These are of economic importance, since diamonds and gold are found in streams that drain the conglomerates.

Thin-section analyses of vertical and horizontal cuts of the Uairen sediments sampled for this study reveal a large amount (between 70 and 90%) of quartz cemented by authigenic sericite and muscovite cemented by sericite. These samples are stained with iron hydroxide and they also show hematite, feldspar, illite, zircon, microcline, apatite, epidote, and magnetite. X-ray diffraction analyses revealed, in decreasing order of concentration, quartz, muscovite, feldspar, hematite, illite, magnetite, zircon, apatite, epidote, chlorite, tourmaline, piemontite, and kaolinite. The presence of muscovite cemented by sericite and minerals such as epidote and piemontite may indicate a low-grade of regional metamorphism (Tucker 1991). X-ray diffraction analyses of the poorly indurated sandstones show quartz, feldspar, muscovite, and a large amount of hematite.

Paleomagnetic results and magnetic petrology

Specimens were stepwise demagnetized using alternating fields (AF) and temperatures. AF demagnetization at 5 mT increments up to 35 or 40 mT was carried out in all samples, but in those cases where NRM intensity remained high at 40 mT further thermal demagnetization was applied. AF and thermal demagnetizations were carried out using a multiaxis tumbler demagnetizer and a standard Schonstedt furnace, respectively. NRM measurements were performed in a parastatic magnetometer, a Digico spinner magnetometer, and a Molspin spinner magnetometer.

Volume magnetic susceptibility was measured after each heating step throughout complete thermal demagnetization to monitor possible chemical changes of the magnetic minerals. These measurements revealed no significant changes in volume susceptibility even at high temperatures.

Isothermal remanent magnetization (IRM) acquisition curves were obtained for some Uairen samples to identify the main magnetic minerals responsible for carrying their remanence

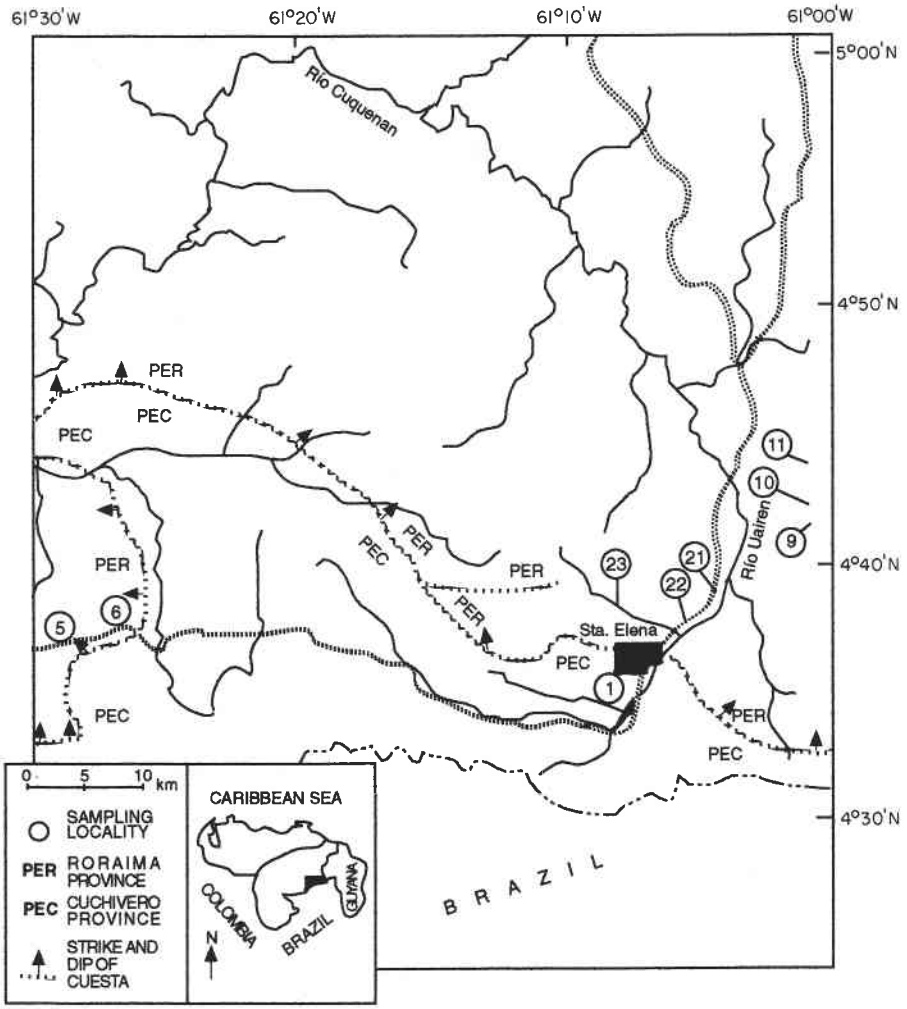


FIG. 1. Sketch map of the Santa Elena area showing sampling localities. Double dotted line represents the main access road to the sampling sites. Inset map of Venezuela shows the area of study in black.

(Fig. 2). Some samples acquired a saturation magnetization at about 0.3 T. This probably indicates the presence of magnetite (Fig. 2B). Some other samples exhibit IRM partial saturation at about 0.3 T; however, the intensity increases progressively thereafter, so complete saturation is not attained at the maximum applied field (Fig. 2A). This behaviour could indicate the presence of two different magnetic minerals in a single sample, namely magnetite and hematite. Most samples, however, did not exhibit saturation, even at 0.65 T, and their IRM curves show a characteristic hematite behaviour (Fig. 2A).

Three different (bedding-corrected) directions of remanence, U1N, U1R, and U2, were isolated by AF and thermal demagnetization. In some cases two of these magnetizations (U2 and either U1N or U1R) were present in the same sample. Figures 3 and 4 show the AF and thermal demagnetization response of the NRM of samples S6B-2 and S11A-2. A soft U2 remanence with coercivities (H_C's) of about 20 mT is stripped away to reveal underlying hard U1N (S6B-2) and U1R (S11A-2) remanences carried by fine-grained magnetite with laboratory unblocking temperatures (T_{UB}'s) between 300 and 580°C (Dunlop 1981, 1983) and hematite unblocking temperatures >600°C. However, most Uairen samples are univectorial, showing only high-H_C and high-T_{UB} components U1N and U1R. Figures 5 and 6 show two examples of AF and thermal cleaning of univectorial samples S23-5, S1B-4, S21-7, and S1A-3. High H_C's and T_{UB}'s > 600°C for U1N (S23-6, S21-7) and U1R (S1A-3, S1B-4) suggest hematite as the main magnetic carrier of these two components.

Components U1N, U1R, and U2 were defined for each sample by linear trajectories on orthogonal vector plots (Zijderveld 1967) and tested by principal-component analyses (Kirschvink 1980). All 22 sites (133 samples) sampled exhibited high-H_C and high-T_{UB} components (either U1N or U1R). Average site means (Tables 1 and 2; Fig. 7) are declination $D = 342°$, inclination $I = 6°$ ($k = 14$, $\alpha_{95} = 10°$, $N = 14$) for U1N and $D = 155°$, $I = 10°$ ($k = 8$, $\alpha_{95} = 17°$, $N = 8$) for U1R. Only six sites exhibited low-H_C and low-T_{UB} components (U2). The average site mean for U2 (Table 3; Fig. 7) is $D = 338°$, $I = -7°$ ($k = 5$, $\alpha_{95} = 25°$, $N = 6$).

Discussion

Figure 7 shows site and grand means for U1N, U1R, and U2 magnetizations. The direction of remanence for U2 closely resembles that for U1N magnetization. However, U2 is much more scattered ($k = 5$) than U1N ($k = 14$). U1R magnetiza-

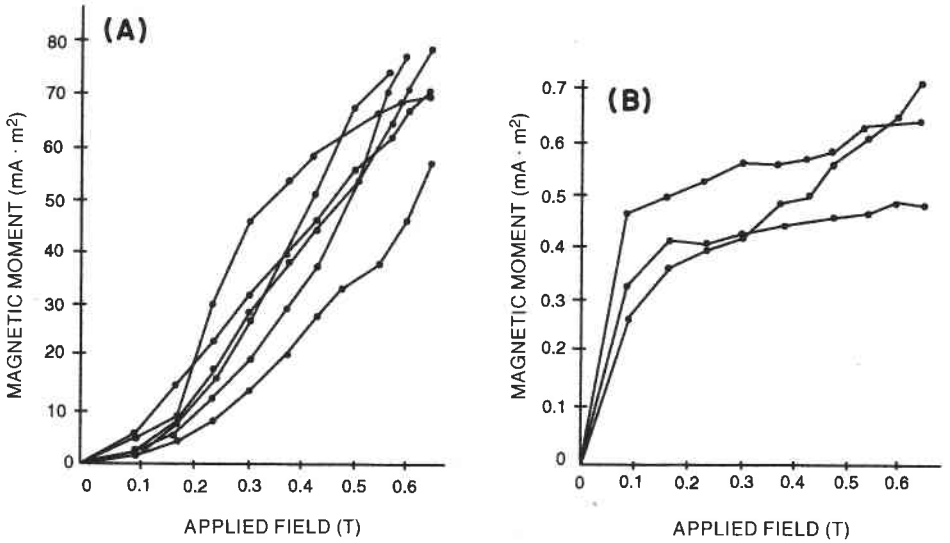


FIG. 2. Saturation isothermal remanent magnetization (SIRM) acquisition curves for Uairen samples. Saturation fields are generally much higher for the univectorial U1N- or U1R-bearing samples (A) than for those carrying U2 (B).

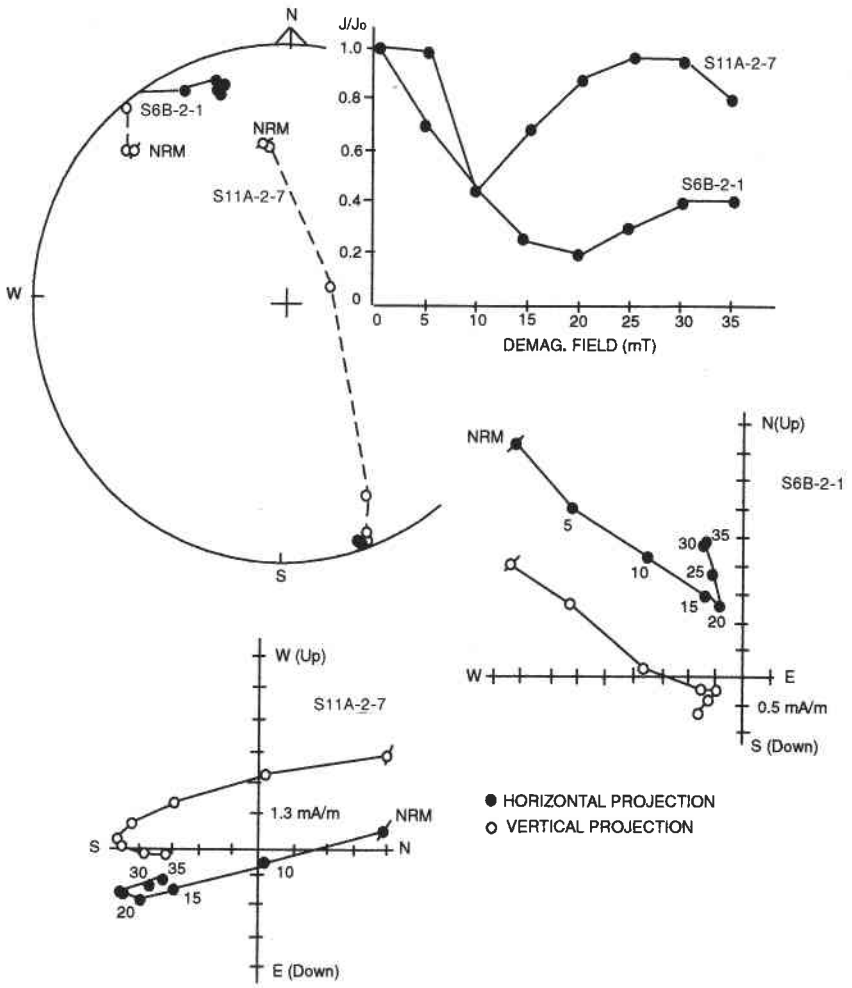


FIG. 3. Typical coercivities of U2, U1N, and U1R magnetizations. Equal-area and orthogonal vector projections of NRM vectors in the course of AF demagnetization, and the corresponding total intensity decay curves are shown for samples S6B-2-1 and S11A-2-7. For samples S6B-2-1 and S11A-2-7, directions of remanence swing from U2 to U1N and from U2 to U1R, respectively. J, magnetization intensity during stepwise AF demagnetization; J_o, initial natural remanent magnetization intensity.

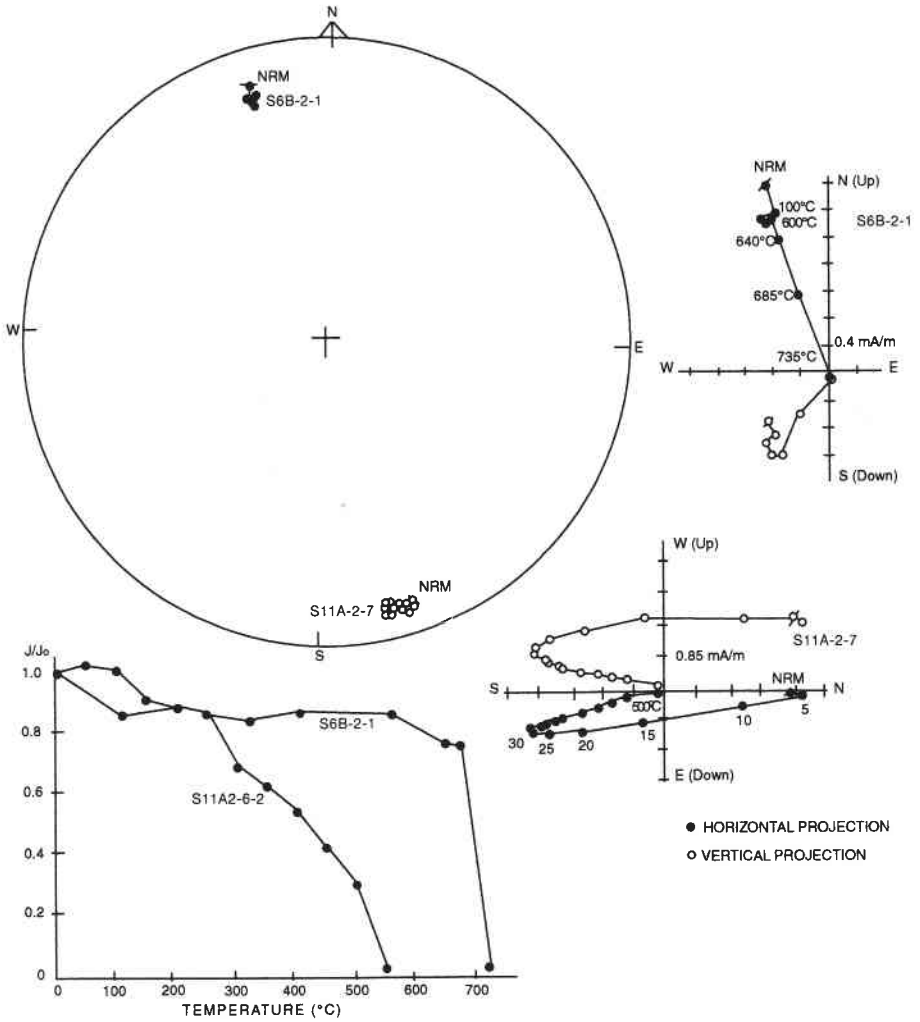


FIG. 4. Further thermal demagnetization of high-T_{UB}'s U1N and U1R magnetizations in samples S6B-2-1 and S11A-2-7. The figure shows U1N and U1R stable end points for samples S6B-2-1 and S11A-2-7, respectively.

TABLE 1. U1N site and grand mean directions

Site	D (°)	I (°)	N	α_{95} (°)	k
5	337	13	9	24	6
6A	340	−12	5	23	13
6B	352	20	4	17	32
6C	291	−6	4	11	78
6D	340	5	8	24	7
6E	357	1	6	25	9
6F	11	15	3	15	71
6G	354	−2	4	24	16
9-1	348	22	7	16	15
9A-2	353	8	1		
10	329	−10	7	8	66
21	328	14	12	7	43
22	356	15	12	17	8
23	328	−5	11	6	72
Mean	342	6	14	10	14

NOTES: D and I, declination and inclination of the remanence direction; N, number of samples; α_{95}, radius of the 95% confidence circle about the mean site direction; k, precision parameter.

tion, with a southeasterly declination and shallow inclination, probably records an incomplete reversal of the geomagnetic field that occurred some time after U1N acquisition.

Hargraves (1968) reports two distinct groups of magnetization for the Roraima Intrusive Suite: a northwesterly positive direction (group I: $D = 333.5°$, $I = 27.5°$, $\alpha_{95} = 11°$) that resembles U1N and U2, and a U1R-like southerly positive direction (group II: $D = 145°$, $I = 39.5°$, $\alpha_{95} = 12°$). From that study he concluded that these magnetizations were acquired either by an incomplete reversal of the Precambrian geomagnetic field or throughout two different times of diabase intrusion. Veldkamp et al. (1971) and Onstott and Hargraves (1981) also report groups I and II for Roraima diabase intrusions in Suriname and Venezuela, respectively. Directional similarities between Uairen components U1N and U2 and their counterparts in the Roraima Intrusive Suite suggest that primary DRM in these sediments has been completely overprinted either by the latter intrusives and (or) by pervasive hydrothermal alteration (Onstott and Hargraves 1981) probably associated to diabase intrusions as well.

However, no evidence for Roraima diabases was observed near the Uairen sites sampled for this study. Although this is not a strong argument against overprints, since the presence of

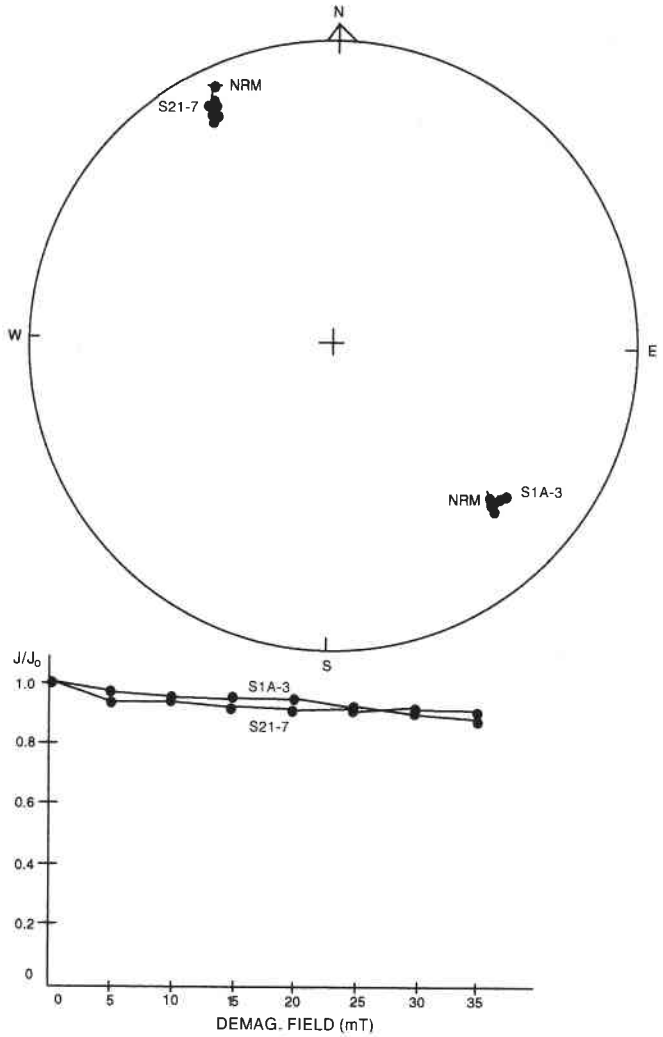


FIG. 5. Coercivities in the course of AF cleaning for U1N and U1R remanences in univectorial samples S21-7 and S1A-3. The figure shows characteristic U1N and U1R directions of remanence for samples S21-7 and S1A-3, respectively.

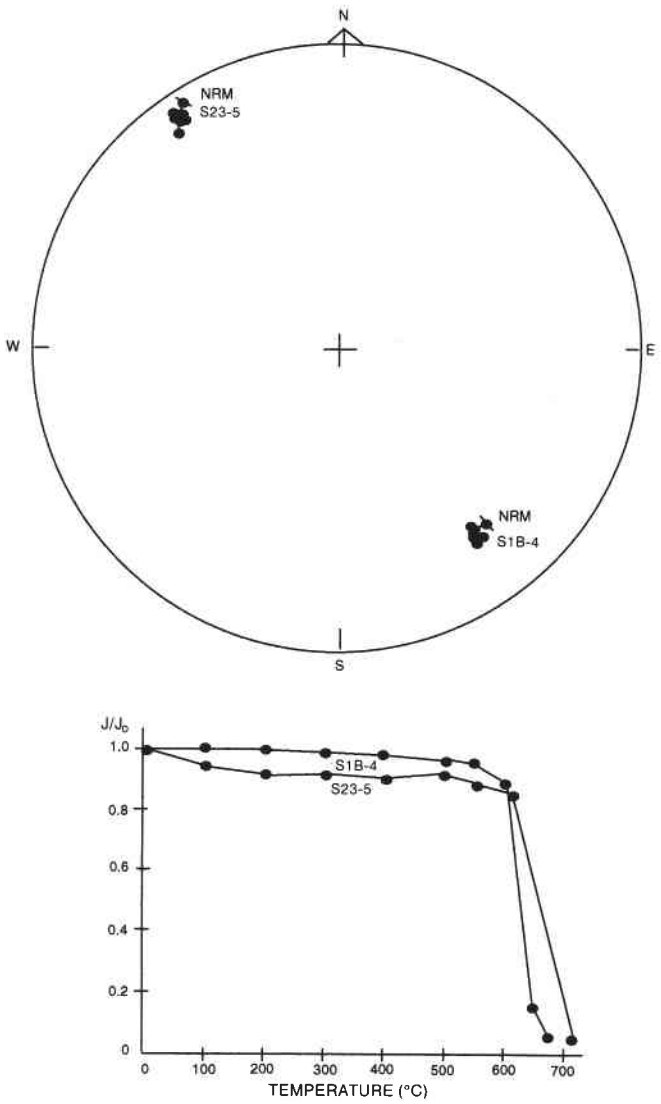


FIG. 6. High-T_{UB}'s U1N and U1R magnetizations in univectorial samples S23-5 and S1B-4. The figure shows U1N and U1R stable end points for samples S23-5 and S1B-4, respectively.

TABLE 2. U1R site and grand mean directions

Site	D (°)	I (°)	N	α_{95} (°)	k
1A	139	24	6	5	186
1B	144	25	7	12	28
1C	147	21	6	4	347
1D	153	33	5	8	107
6H	178	−33	6	6	128
9A1	157	−2	1		
11A1	174	31	1		
11A2	150	−24	1		
Mean	155	10	14	17	8

TABLE 3. U2 site and grand mean directions

Site	D (°)	I (°)	N	α_{95} (°)	k
6A	14	−12	5	29	10
6B	303	−26	4	24	144
6D	313	19	9	10	28
9A2	353	3	1		
11A2	319	−37	4	30	10
22	1	−15	12	27	13
Mean	338	−7	6	25	4

overlying sills now eroded would be unobserved in outcrop, these intrusives seem to be concentrated (independently of the degree of erosion) at the northern end of the Roraima Group, far away from the Santa Elena area, and where a major zone of weakness has been hypothesized by Bellizia (1957). The low metamorphism of Uairen samples, revealed by thin-section analyses, argues against total thermal overprint as well.

To harmonize these observations with paleomagnetic and rock magnetic results we propose here an alternative explanation for the origin of U1N, U1R, and U2 magnetizations. The

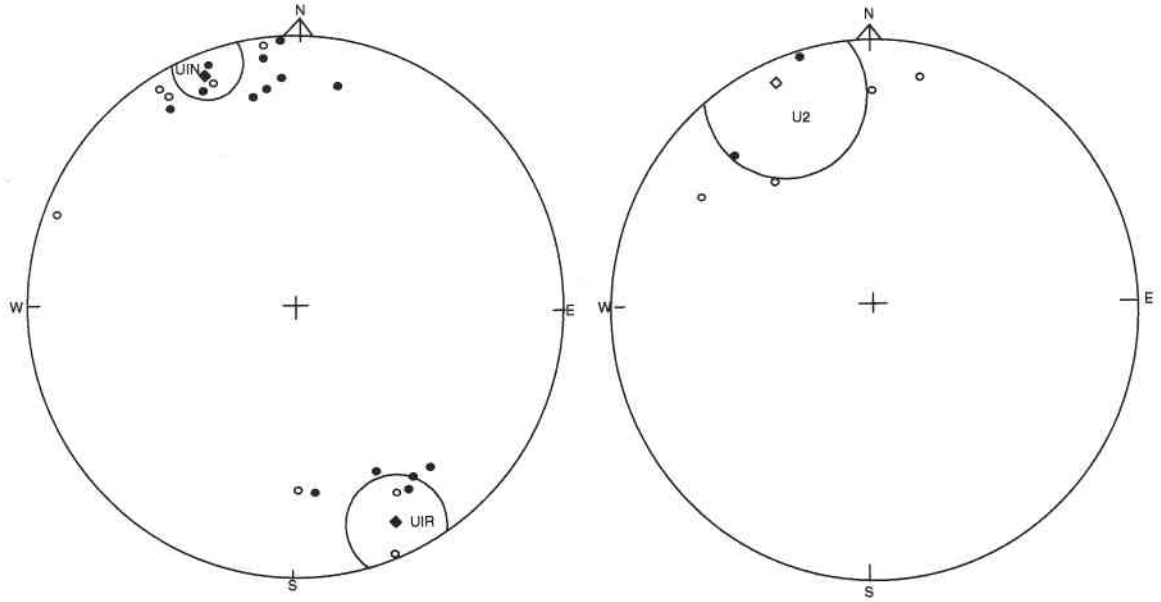


FIG. 7. Equal-area projections of site mean U1N, U1R, and U2 directions together with grand means and circles of 95% confidence. ◇, final means.

coexistence of an unstable and a stable component with directional similarities in some Uairen samples suggests a two-step scenario for NRM acquisition. Less stable U2 was perhaps the first magnetization acquired as a DRM. However, since U2 is carried by low-H_C (20 mT) multidomain magnetite (Bailey and Dunlop 1983; Dunlop 1983), it is reasonable to expect that magnetic torque would be outweighed by its mechanical counterpart (King 1955), resulting in the directional scattering that characterizes this component. Figure 7 shows that such a scattering is more evident in the declination than in the inclination values of U2 site means. Randomness in declination values is the expected result of misorientation by mechanical torque when elongated grains of magnetic minerals with sizes over 1 μm touch bottom (King 1955). Scatter could also be attributed to apparent polar wander (APW) recorded by different site means corresponding to different sedimentary horizons. However, APW does not explain the grain-size dependence that seems to relate U2 NRM to multidomain magnetite.

For the fine-grained magnetite, magnetic torque should outweigh mechanical misaligning effects. Hence, U1N shows a much better grouping than U2. U1N is carried not only by fine-grained magnetite but mostly by hematite (e.g., Fig. 6), a byproduct of magnetite oxidation during low-temperature diagenetic processes which, judging by the directional similarity of U1N to U2, should be penecontemporaneous with deposition and consolidation of the sedimentary layer. Therefore U1 thermochemical remanent magnetization (TCRM) overprint was probably acquired within a short time interval of no more than 10^5 years (Baag and Helsley 1974; Herrero-Bervera and Helsley 1983; Shive et al. 1984), which included an incomplete reversal of the geomagnetic field (U1R). Fine-grained magnetite oxidation could also date from later hydrothermal alteration associated with the emplacement of the Roraima intrusives. This possibility seems unlikely though, not only because of the evidence provided by thin-section analyses but also because it fails to explain why an apparently unique direction of remanence U1 is indistinctively carried by both unaltered primary fine-grained magnetite and its secondary and much younger oxidation byproduct.

Uairen sediments lack fossils and authigenic dateable minerals; therefore there is no way of obtaining a precise age for U1N, U1R, and U2. Radiometric ages of Cuchivero volcanics and Uaimaipue pyroclastics are, respectively, the oldest and youngest time constraints available for the Uairen Formation. Thus, providing that U1N, U1R, and U2 indeed date from the time of deposition, consolidation, and diagenesis of these sediments, all we can conclude is that they should have ages no older than 1960 Ma and no younger than 1580 Ma. In any case this is the first evidence ever reported for primary NRM's preserved in Middle Proterozoic sedimentary rocks (Henthorn 1981; Zhang and Zhang 1985).

Conclusions

The following conclusions can be drawn from this study:

(1) Three directions of remanence, namely U1N, U1R, and U2, have been isolated from the Uairen Formation, the basal and oldest sedimentary sequence of the Roraima Group. U1N and U1R are carried by fine-grained magnetite and hematite, whereas U2 resides in coarse-grained magnetite.

(2) Two hypotheses to explain the origin of these NRM's have been discussed here. The first hypothesis proposes a complete thermal overprint of Uairen sediments by later intrusions of diabases (Roraima Intrusive Suite). The second hypothesis suggests U2 is a primary DRM dating from the time of deposition and U1N and U1R are TCRM's penecontemporaneous with deposition, and acquired during diagenesis and consolidation of the sedimentary column. The directional scatter of U2 can be attributed to mechanical misaligning effects on large magnetite crystals at the time of DRM acquisition.

(3) Evidence for the first hypothesis includes the presence of Uairen-like directions of remanence (groups I and II) as characteristic components of the Roraima Intrusive Suite. Evidence for the second hypothesis includes the apparent absence of diabase dikes at and near the sampled localities and the low

metamorphism revealed by thin-section analyses of the Uairen samples.

(4) Providing this second hypothesis is correct and according to the time constraints set by the radiometric ages available for the basement volcanics (Cuchivero) and the Uaimaipue pyroclastics in the Santa Elena area, it is reasonable to conclude that the beginning of Uairen deposition and the acquisition of U1N, U1R, and U2 magnetizations occurred sometime between 1580 and 1960 Ma. Thus U2 would be one of the oldest DRM's ever reported.

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